

# Accessible Documents in LaTeX

[https://github.com/rhstanton/accessible\\_LaTeX](https://github.com/rhstanton/accessible_LaTeX)

Version 1.5

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## 1 Introduction

On January 8, 2026, we were notified by campus that, beginning on April 24, 2026:

“The updated requirements of the ADA require that digital course materials provided to students, even materials inside password-protected course sites like bCourses, will need to comply with accessibility standards (Web Content Accessibility Guidelines (WCAG) 2.1 Level AA).”

Many of us use L<sup>A</sup>T<sub>E</sub>X to create teaching materials—both slides and documents. Standard L<sup>A</sup>T<sub>E</sub>X (including Beamer) does not automatically generate accessible PDFs.

## 2 What is this project?

An experiment exploring accessible L<sup>A</sup>T<sub>E</sub>X documents using the L<sup>A</sup>T<sub>E</sub>X Tagging Project:

- **Two templates:** Articles (`article` class) and slides (`ltx-talk`)
- **Shows:** What’s common to both, what’s different
- **Contains:** Working examples with math, text, graphics, and tables
- **Scores:** Perfect 100% from the bCourses accessibility checker (Ally)

### 2.1 How to use it

1. **Learn the common requirements:** Apply to any L<sup>A</sup>T<sub>E</sub>X document
2. **See class-specific needs:** Understand slides vs. articles
3. **Copy and adapt:** Use as templates for your documents
4. **Study the code:** Heavily commented for learning

Available at: [https://github.com/rhstanton/accessible\\_LaTeX](https://github.com/rhstanton/accessible_LaTeX)

## 2.2 Converting to accessible LaTeX: Overview

- **Your existing L<sup>A</sup>T<sub>E</sub>X skills transfer** - you're just adding accessibility features
- **The changes are minimal** and follow consistent patterns

**Common requirements** (both slides & articles):

- Add document metadata and enable PDF tagging
- Tag images with alt text; tag table headers
- Use accessible colors; compile with LuaLaTeX

**Key difference:**

- Articles: **Keep your existing class!** Just add accessibility features
- Slides: Switch to `ltx-talk` (similar syntax, but recreate styling)

Both require LaTeX kernel 2025-11-01 or later (see next sections for details).

See “Getting started” sections at the end for detailed steps. For complete LaTeX Tagging Project documentation, see [latest detailed instructions](#).

## 3 TeX version requirements (both slides and articles)

- **Minimum:** TeX Live 2023 or later with all packages updated
- **Critical:** Must have LaTeX kernel 2025-11-01 or later
  - Update via TeX Live Manager (Windows) or TeX Live Utility (Mac)
  - Or use Overleaf Labs (see next section)
- **Will NOT work:** TeX Live 2022 or earlier

## 4 You CAN use Overleaf (both slides and articles)

- `ltx-talk` requires a very recent **LaTeX kernel** (2025-11-01 or later)
- This is available through Overleaf’s [Labs program](#) (not standard Overleaf)

### 4.1 Using Overleaf (2 steps)

#### 1. Join Overleaf Labs:

- Visit <https://www.overleaf.com/labs/participate>
- Opt in and enable “Rolling TeX Live releases”

#### 2. Configure project:

- Set TeX Live version to “**Rolling TeXLive (labs)**” (bottom of list)
- Set Compiler to **LuaLaTeX**

**Resources:** <https://docs.overleaf.com/writing-and-editing/creating-accessible-pdfs>

## 5 Common requirements for ALL documents

Whether you’re creating slides or articles, these requirements are [the same](#):

### 5.1 Document Metadata

- Every accessible document **must** begin with `\DocumentMetadata`
- Goes *before* `\documentclass`

```
\DocumentMetadata{
  pdfstandard=a-2u,      % PDF/A-2u format
  pdfstandard=ua-1,      % Add PDF/UA-1 conformance target
  pdfversion=1.7,        % Explicit PDF version
  lang=en-US,            % Language for screen readers
  tagging=on,            % Enable PDF tagging
  tagging-setup={
    math/alt/use,         % Keep math accessibility enabled (screen readers get tagged math)
    math/mathml/AF=false, % Do not embed MathML as PDF Associated Files
    math/tex/AF=false,    % Do not embed TeX source as Associated Files
    math/mathml/sources= % Clear MathML source attachment list
  }
}
```

This configures accessibility, math tagging, and validator-compatibility behavior in one place. The `tagging-setup` entries keep math accessible while reducing helper attachments that can trigger warnings in some strict conformance validators.

### 5.2 Images, Tables, and Colors

- **Images:** All images must include alt text descriptions
  - Allows screen readers to describe visual content
  - Even decorative images need marking
  - See dedicated “Figures” section for examples
- **Tables:** Must specify which rows are headers
  - Helps screen readers navigate table structure
  - Essential for data accessibility
  - See dedicated “Tables” section for examples
- **Accessible colors:** WCAG 2.1 requires 4.5:1 contrast ratio

```
\colorlet{AccessibleRed}{red!80!black}
\colorlet{AccessibleGreen}{green!40!black}
% Standard blue is fine as-is
```

### 5.3 LuaLaTeX Compilation

- You **must** compile with LuaLaTeX (not pdfLaTeX or XeLaTeX)
- **Why LuaLaTeX?**
  1. **Automatic MathML:** Makes math accessible without manual work
  2. **Full tagging support:** Complete PDF accessibility features
  3. **Modern fonts:** Handles Unicode and OpenType fonts
- **How to switch:**
  - Command line: `lualatex filename.tex`
  - Most editors: Select “LuaLaTeX” from compiler menu

## 6 The basics

- Use standard L<sup>A</sup>T<sub>E</sub>X environments: `section`, `subsection`, `itemize`, `enumerate`, etc.
- **Existing source files don’t need a lot of editing**
- Here’s some gratuitous *math* for the accessibility checker

## 7 Figures

When including a figure, you **must provide alt text** describing the image for screen readers. This is *required* for accessibility compliance.

```
\includegraphics[height=.4\textheight,alt={A capybara}]{capybara.jpg}
```



Figure 1: A picture of a capybara

## 8 Tables

When including a table, you **must specify which rows are headers**. Screen readers use this information to help users navigate table content. Use `\tagpdfsetup{table/header-rows={...}}` *before* the `tabular` environment:

- Use `{1}` for 1 header row
- Use `{1,2}` for 2 header rows
- Use `{1,2,3}` for 3 header rows, etc.
- **Validator note:** PAC can sometimes report Table header cell has no associated subcells even when headers are tagged correctly. The LaTeX team documents this behavior at <https://github.com/latex3/tagging>

### 8.1 Example: Table with 3 Header Rows

```
\tagpdfsetup{table/header-rows={1,2,3}}
\begin{tabular}{cccccccr}
\toprule
← 3 header rows
\midrule
← data rows
\bottomrule
\end{tabular}
```

| Payment<br>date | Caplet<br>expiry date | $DF_{\text{pay}}$ | Forward<br>rate | Days<br>to<br>expiry | Days in<br>accrual<br>period | $T_{\text{expiry}}$ | $\Delta$ | Caplet    |
|-----------------|-----------------------|-------------------|-----------------|----------------------|------------------------------|---------------------|----------|-----------|
| 2004/12/01      | —                     | 0.99550           | 0.01790         | 0                    | 91                           | 0.00000             | 0.25278  | —         |
| 2005/03/01      | 2004/11/29            | 0.99008           | 0.02188         | 89                   | 90                           | 0.24384             | 0.25000  | 1,178.77  |
| 2005/06/01      | 2005/02/25            | 0.98401           | 0.02413         | 177                  | 92                           | 0.48493             | 0.25556  | 4,844.73  |
| 2005/09/01      | 2005/05/27            | 0.97733           | 0.02675         | 268                  | 92                           | 0.73425             | 0.25556  | 10,016.71 |

Table 1: A table

## 9 Common pitfalls

- **Confusing PDF tagging with PDF bookmarks**
  - `tagging=on` creates structure tree for **screen readers** (required for accessibility!)
  - PDF bookmarks (navigation outline in left pane) are **separate and optional**
  - Bookmarks do NOT generate automatically from `tagging=on`
  - For articles: use `\tableofcontents` or the `bookmark` package
  - For slides: See custom bookmark code in `accessible_slides.tex` preamble (Part 2)
- **Forgetting alt text for images**
  - Every `\includegraphics` needs an `alt={...}` parameter
  - For purely decorative images, use empty alt text: `alt={}`
- **Not specifying table header rows**
  - Add `\tagpdfsetup{table/header-rows={...}}` before each table
  - Use `{1}` for 1 header row, `{1,2}` for 2 header rows, etc.

- **Insufficient color contrast**
  - WCAG 2.1 requires 4.5:1 contrast ratio for normal text
  - Avoid light colors: **yellow**, **cyan** fail contrast requirements
  - Darken **red** and **green**: use **red!80!black**, **green!40!black**
  - Standard **blue** is fine and meets WCAG requirements
  - Test with a contrast checker: <https://webaim.org/resources/contrastchecker/>
- **Using the wrong compiler**
  - Make sure your editor is set to use **LuaLaTeX**, not **pdfLaTeX**
- **Old TeX distribution**
  - TeX Live 2022 or earlier won't work
  - Update packages using TeX Live Manager (Windows) or TeX Live Utility (Mac)

## 10 What to ADD to existing documents

When converting existing LaTeX files to be accessible, here's what you need to add:

### 10.1 Required additions for ALL documents

1. **At the very top** (before `\documentclass`):

```
\DocumentMetadata{
  pdfstandard=a-2u,
  pdfstandard=ua-1,
  pdfversion=1.7,
  lang=en-US,
  tagging=on,
  tagging-setup={...see template...}
}
```

2. **In every `\includegraphics` command:**

```
\includegraphics[alt={Description of image}]{file}
```

3. **Before every table:**

```
\tagpdfsetup{table/header-rows={1}}
\begin{tabular}{...}
```

4. **Change your compiler to LuaLaTeX** (not pdfLaTeX or XeLaTeX)

### 10.2 Additional for existing Beamer slides

- Change `\documentclass{beamer}` to `\documentclass[frame-title-arg]{ltx-talk}`
- Remove all Beamer themes, color schemes, and template commands
- Keep your `\begin{frame}` environments as-is
- Recreate styling using standard LaTeX packages (xcolor, fontspec, etc.)
- This is **one-time preamble work**—then reuse for all future talks!

## 11 Getting started: Common steps for both

1. **Setup environment:** Use Overleaf Labs OR install/update TeX Live locally
  - Overleaf: See earlier section for Labs setup
  - Local: Update via TeX Live Manager (Windows) or TeX Live Utility (Mac)
2. **Get templates:** Download from [https://github.com/rhstanton/accessible\\_LaTeX](https://github.com/rhstanton/accessible_LaTeX)
3. **Add accessibility features:**
  - Add `alt` text to images
  - Add `table/header-rows` to tables
4. **Set compiler to LuaLaTeX**
5. **Compile:**
  - Compile with LuaLaTeX

## 12 Getting started: What's different by document type

### 12.1 For Articles (using `article`, `report`, `book`, etc.)

- Add `\DocumentMetadata` before `\documentclass`
- Include the `tagging-setup` block in `\DocumentMetadata` (as shown in this template)
- Switch to `fontspec` and `unicode-math` packages
- **Keep your existing `documentclass`!**

### 12.2 For Slides (migrating from Beamer)

- Copy preamble from `accessible_slides.tex`
- Change `\documentclass{beamer}` to `\documentclass[frame-title-arg]{ltx-talk}`
- **Remove Beamer themes/templates/colors**
- **Recreate styling** using standard LaTeX/xcolor
- **One-time preamble work**—then reuse for all future talks!

## 13 Validation workflow: use multiple checkers

- No single checker catches everything.
- Accessibility usability checks and strict PDF conformance checks are not identical.
- A practical workflow is to run more than one checker and compare findings.

**Checkers used so far:** Ally (the accessibility checker built into bCourses), veraPDF (open-source PDF conformance validator, <https://verapdf.org/>), PAC (PDF Accessibility Checker, <https://pac.pdf-accessibility.org/>), and Acrobat.

## 14 Optional validator cleanup step: `strip_af.py`

- The metadata settings alone are enough to generate a 100% score on Ally.
- Stricter checkers (e.g., veraPDF) may still complain about /AF or embedded-file structures.
- Optional strict-archival cleanup command:

```
python strip_af.py build/yourfile.pdf
```

- Then validate `build/yourfile.pdf` with veraPDF.
- Prerequisite for the script: `python -m pip install pikepdf`

## 15 Validator-specific notes

- Not all validator warnings indicate real source problems.
- Acrobat: “Lbl and LBody – Failed”
  - Often appears around figure captions, section numbers, and table captions
  - This is usually an Acrobat checker bug, not a source-tagging error
  - The `<Lbl>` tag is valid for numbering
  - Reference: <https://tex.stackexchange.com/a/709655/2388>
- PAC table warning (known behavior)
  - PAC can report `Table header cell has no associated subcells` even when `table/header-rows` is correct
  - The LaTeX Tagging Project documents this behavior at <https://github.com/latex3/tagging-project/issues/1056>
- When this warning appears
  - First confirm source markup is correct (especially `table/header-rows`)
  - Keep source tagging and document the validator warning
  - Report minimal reproductions to tool vendors when possible
- **Interpretation tip:** Treat one-tool warnings as signals to investigate, not automatic proof of source errors.

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